

BF29.5

1. Since $\tau = \frac{L}{R_{eq}}$ for an LR circuit, $\tau_b = \tau_c > \tau_a$ because $2\frac{L}{R} = 2\frac{L}{R} > \frac{L}{R}$.
2. Knowns: $I(t)$, $N = 400$, $r = 0.02$ m, $l = 0.2$ m, $d = 0.01$ m
Unknowns: $E(t) = ?$

$$\begin{aligned}
 -L \frac{dI}{dt} &= \varepsilon = \oint \vec{E} \cdot d\vec{l} \\
 -\left(\frac{\mu_{core} N^2 A}{l}\right) \frac{dI}{dt} &= 2\pi r E \\
 |E| &= \frac{dI}{dt} \frac{\mu N^2 A}{2\pi l r} \\
 &= \frac{dI}{dt} \frac{(1.26 \cdot 10^{-6} \text{ H/m})(400)^2 (\pi(0.02 \text{ m})^2)}{2\pi(0.2 \text{ m})(0.03 \text{ m})} \\
 &= \frac{dI}{dt} (0.00672)
 \end{aligned}$$

This corresponds to graph B.

3. Knowns: $E = 2$ mV/m, $d = 1.5$ m
Unknowns: $dB/dt = ?$

$$\begin{aligned}
 -\frac{d\Phi_b}{dt} &= \oint \vec{E} \cdot d\vec{l} \\
 \frac{dB}{dt} A &= 2\pi r E \\
 \frac{dB}{dt} (\pi(1.5 \text{ m})^2) &= 2\pi(1.5 \text{ m})(0.002 \text{ V/m}) \\
 \frac{dB}{dt} &= \boxed{2.67 \text{ mT/s}}
 \end{aligned}$$

4. Knowns: $L = 12$ mH, $\Delta A = -50$ mA, $\Delta t = 13$ μ s
Unknowns: $\Delta V = ?$, *direction*

$$\begin{aligned}
 \varepsilon &= -L \frac{dI}{dt} \\
 &= -(12 \text{ mH}) \frac{-50 \text{ mA}}{13 \mu\text{s}} \\
 &= \boxed{46 \text{ V}}
 \end{aligned}$$

Since ΔV is positive, the voltage is increasing in the direction of current.

5. Knowns: $r = 0.0125$ m, $l = 0.14$ m, $N = 180$, $I = 0.8$ A
Unknowns: $U_l = ?$

$$\begin{aligned}
 U_l &= \frac{1}{2}LI^2 \\
 L &= \frac{\mu N^2 A}{l} \\
 &= \frac{(1.26 \cdot 10^{-6} \text{ H/m})(180)^2(\pi(0.0125 \text{ m})^2)}{0.14 \text{ m}} \\
 &= 0.000143139 \text{ H} \\
 U_l &= \frac{1}{2}(0.000143139 \text{ H})(0.8 \text{ A})^2 \\
 &= \boxed{0.0000458 \text{ J}}
 \end{aligned}$$

6. Knowns: $L = 3 \text{ mH}$, $C_{\min} = 160 \text{ pF}$, $C_{\max} = 180 \text{ pF}$

Unknowns: $f_{\min} = ?$, $f_{\max} = ?$

$$\begin{aligned}
 \omega &= \frac{1}{\sqrt{LC}} \\
 f &= \frac{\omega}{2\pi} \\
 f &= \frac{1}{2\pi\sqrt{LC}} \\
 f_{\max} &= \frac{1}{2\pi\sqrt{(3 \text{ mH})(160 \text{ pF})}} \\
 &= 229720 \text{ Hz} \\
 f_{\min} &= \frac{1}{2\pi\sqrt{(3 \text{ mH})(180 \text{ pF})}} \\
 &= 216582 \text{ Hz}
 \end{aligned}$$

7. Knowns: diagram

Unknowns: $I_i = ?$, $I_f = ?$

Initially, the inductor takes all of the voltage from the battery on its branch. Thus, the only branch with current passing through it is the one resistor branch, so

$I = \frac{10 \text{ V}}{20 \Omega} = 0.5 \text{ A}$. After some time, both branches have current passing through them and the inductor acts as a normal wire, so $I = \frac{10 \text{ V}}{10 \Omega} = 1 \text{ A}$.

8. Knowns: diagram

Unknowns: $\Delta V_{1200} = ?$

$$\begin{aligned}
 U_c &= U_c \\
 \frac{1}{2}CV_1^2 &= \frac{1}{2}CV_2^2 \\
 (300 \mu\text{F})(140 \text{ V})^2 &= (1200 \mu\text{F})V_2^2 \\
 V_2 &= \boxed{70 \text{ V}}
 \end{aligned}$$

9. Knowns: diagram

Unknowns: $I_1 = ?$, $I_2 = ?$, $I_3 = ?$

Initially, the inductor acts as a resistor with infinite resistance. In this instant, it's as if

there is a 30 ohm resistor in series with the battery, so there is $\boxed{1 \text{ A}}$ of current through R20. After a long time, the inductor is fully charged and the resistor is shorted out of the circuit, so there are $\boxed{0 \text{ A}}$ of current through R20. After the switch is reopened, the current remains at $\boxed{3 \text{ A}}$, because the inductor can't change current instantly.

10. Since $\tau = \frac{L}{R}$, we can write the equation $I(t) = I_0(1 - e^{-t/\tau})$. Since $V = IR$, we know that $V(t) = V_0(1 - e^{-t/\tau})$. We can rearrange this into the following equation to know what to graph: $L = -\frac{Rt}{\ln\left(1 - \frac{V(t)}{V_0}\right)}$. Thus, we can graph t and $\ln(V(t))$. By our equation, $L = 3 \text{ mH}$.